



Department of Software Engineering

Database System

Project report

One Tag One Nation

Submitted to

Maam Saima

Submitted by

Maryam shakil(24-SE-01)

Ayesha waqar(24-SE-79)

TOLL MANAGEMENT SYSTEM

(User & Admin Portal)

1. Introduction:

The Toll Management System is a desktop-based application developed using Python and Oracle Database. It automates toll road operations by managing users, vehicles, RFID tags, toll transactions, and system administration.

The system is divided into two major modules:

- User Portal (Customer Side)
- Admin Panel (Management Side)

2. Objectives

- Provide secure login system for users and admin
- Manage vehicles and RFID tags efficiently
- Enable automatic toll deduction tracking
- Provide wallet recharge system
- Allow admin to manage full system data
- Improve transparency and automation in toll system

3. System Overview

The system consists of two roles:

- **User Role:**
 - ✓ Manage profile
 - ✓ Add/view vehicles
 - ✓ Manage RFID tags
 - ✓ Recharge wallet
 - ✓ View toll history
 - ✓ Send feedback
- **Admin Role:**
 - ✓ Manage users
 - ✓ Manage vehicles
 - ✓ Manage RFID tags
 - ✓ Control toll plazas
 - ✓ Manage motorways
 - ✓ View reports

- ✓ Monitor transactions
- ✓ Handle feedback

4. USER PORTAL MODULES

4.1 Authentication Module

Users can register and login securely using username and password.

ONE TAG ONE NATION — Login

ONE TAG ONE NATION

Toll Management System — Pakistan

☒ Login ☐ Sign Up

Select Portal: ☒ User ☐ Admin

Username

Enter username

Password

Enter password

Login →

• DB Connected | localhost:1521/ord1

Demo — User: maryam / 123 Admin: admin / admin123

4.2 User Dashboard

After login, the user is directed to a dashboard containing only essential features.

The dashboard for Maryam Shakil (ID: 1) shows a welcome message and a status bar indicating a connection to localhost:1521/ord1. It features four summary cards: My Vehicles (2), Active Tags (2), Total Trips (2), and Total Paid (Rs 238). Below these are two tables: MY TAG BALANCES and RECENT TRANSACTIONS.

TagID	Plate	Vehicle Class	Balance (Rs)	Reserve Bal	Status	Expiry
1001	LEA-1234	Car	4887.5	500.0	ACTIVE	07-Jun-31
1003	RWP-9012	Truck 2 axle	3000.0	300.0	ACTIVE	07-Jun-31

TxnID	Plate	Plaza	Distance(Km)	Total(Rs)	Date
21	LEA-1234	Islamabad Toll Plaza	25.0	112.5	09-Jun-26 15:56
1	LEA-1234	Islamabad Toll Plaza	25.0	125.0	07-Jun-26 13:17

4.3 Profile Module

Users can view and update their personal information.

The MY PROFILE page displays account details for Maryam Shakil. It includes sections for Account Information and Change Password.

Account Information

Name: Maryam Shakil	CNIC: 37485-1234567-1
Phone: 03001234567	Email: maryam@gmail.com
Province: Punjab	Address: Taxila
Emergency Contact: 03009998888	Username: maryam

Change Password

Current Password: New Password: Confirm Password:

[Change Password](#)

4.4 Vehicle Management Module

Users can view vehicles

 MY VEHICLES Vehicles registered under your account							
• Connected — localhost:1521/ord1							
• Registered Vehicles							
VehicleID	Plate	Model	Company	Color	Type	Driver CNIC	Blacklisted
103	RWP-9012	Truck	Hino	Red	Truck 2 Axle	—	N
101	LEA-1234	Corolla	Toyota	White	Car	37405-1111111-1	N

4.5 RFID Tag Module

Users can view RFID tags linked with vehicles.

 MY TAGS RFID tags linked to your vehicles							
• Connected — localhost:1521/ord1							
• Tag Details							
TagID	Plate	Balance (Rs)	Reserve Bal	Status	Issue Date	Expiry	
1003	RWP-9012	3000.0	300.0	ACTIVE	07-Jun-26	07-Jun-31	
1001	LEA-1234	4887.5	500.0	ACTIVE	07-Jun-26	07-Jun-31	

4.6 Wallet / Recharge Module

Users can:
 Recharge balance
 View current balance

RECHARGE MY TAG

Top up your RFID tag balance instantly

Connected

localhost:1521/ord1

Select your tag, enter an amount, choose a payment method and click Recharge.

Recharge Form

Tag: 1001:LEA-1234 (Balance: 5387.5)

e.g. 500

Payment Method: Cash

Bank / TXN Ref: Optional reference no.

Rs 200

Rs 500

Rs 1,000

Rs 2,000

Recharge Now

My Current Tag Balances

TagID	Plate	Balance (Rs)	Reserve Bal	Status	Expiry
1001	LEA-1234	5387.5	500.0	ACTIVE	07-Jun-31
1003	RWP-9012	3000.0	300.0	ACTIVE	07-Jun-31

My Recharge History

RechargeID	TagID	Plate	Amount (Rs)	Method	Bank Ref	Date	Status
23	1001	LEA-1234	500.0	JazzCash	—	09-Jun-26 12:40	SUCCESS
2	1003	RWP-9012	2000.0	JazzCash	JC-002	07-Jun-26 13:17	SUCCESS
1	1001	LEA-1234	3000.0	Bank Transfer	BT-001	07-Jun-26 13:17	SUCCESS

4.7 Toll Transaction and History Module

Displays history of toll deductions with date and amount.

MY TRANSACTIONS

All toll deductions for your account

Connected

localhost:1521/ord1

Transaction History

TxnID	Plate	Plaza	Entry Terminal	Exit Terminal	Distance(Km)	Rate/Km	Base Toll	Surcharge	Discount	Total(Rs)	Date
21	LEA-1234	Islamabad Toll Plaza	Islamabad Entry	Islamabad Exit	25.0	5.0	125.0	50.0	62.5	112.5	08-Jun-26 21:56
1	LEA-1234	Islamabad Toll Plaza	Islamabad Entry	Islamabad Exit	25.0	5.0	125.0	0.0	0.0	125.0	07-Jun-26 13:17

Total Trips: 2

Total Paid: Rs 237.50

Recharge History

RECHARGE HISTORY							
All top-ups applied to your tags							
• Connected — localhost:1521/ord1							
• Recharge Records							
RechargeID	TagID	Plate	Amount (Rs)	Method	Bank Ref	Date	Status
2	1003	RWP-9012	2000.0	JazzCash	JC-002	07-Jun-26 13:17	SUCCESS
1	1001	LEA-1234	3000.0	Bank Transfer	BT-001	07-Jun-26 13:17	SUCCESS

Payment History

PAYMENT HISTORY						
All payment records for your tags						
• Connected — localhost:1521/ord1						
• Payment Records						
PaymentID	Plate	Amount (Rs)	Method	Bank Ref	Date	Status
1	LEA-1234	2000.0	JazzCash	JC-REF-001	07-Jun-26 13:17	SUCCESS

4.8 Status Module

User can see their status of alerts and exemptions

MY ALERTS				
System notifications for your tags				
• Connected — localhost:1521/ord1				
• Notifications				
AlertID	TagID	Plate	Message	Date
2	1003	RWP-9012	Tag Expiring Soon	07-Jun-26 13:17
1	1001	LEA-1234	Low Balance Alert	07-Jun-26 13:17

MY EXEMPTIONS							
Toll discount exemptions on your account							
• Connected — localhost:1521/ord1							
• Active Exemptions							
ExemptionID	Plaza	Reason	Discount Type	Discount Value	Start Date	End Date	Status
1	Islamabad Toll Plaza	Government Official	PERCENT	50	01-Jan-26	31-Dec-26	ACTIVE

4.9 Feedback Module

Users can send complaints or suggestions.

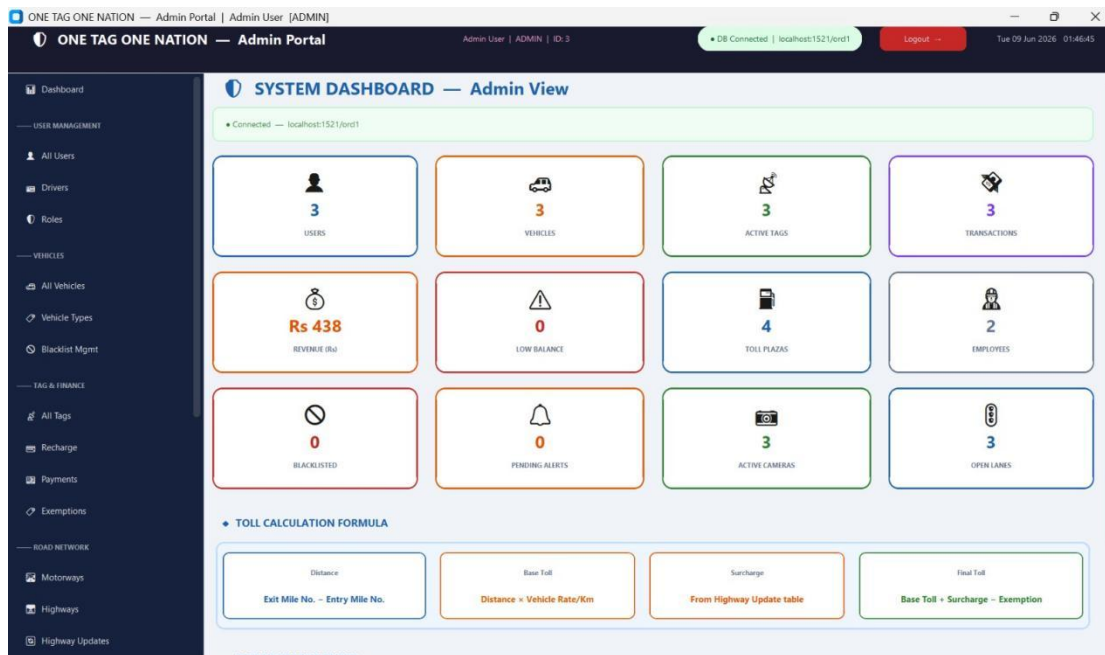
FEEDBACK		
Share your experience with us		
• Connected — localhost:1521/ord1		
• Submit New Feedback		
Rating:	☐ ★ 1 ☐ ★★ 2 ☐ ★★★ 3 ☐ ★★★★ 4 ☒ ★★★★★ 5	
Comments:		
	Submit Feedback	
• My Previous Feedback		
FeedbackID	Rating	Comments
3	5	smooth working
1	5	Excellent Toll Service

Refresh

5. ADMIN PANEL MODULES

5.1 Admin Dashboard

Admin dashboard provides full system control and analytics overview.



5.2 User Management Module

Admin can:

View all users

Manage user accounts

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2020 01:48:46

USER MANAGEMENT

Add, edit and manage all system users

Connected — localhost:1521/orcl1

Add New User

Name: Full Name CNIC: XXXXX-XXXXXXX-X Phone: 0300-XXXXXXX Email: user@email1.com

Province: e.g.- Punjab Street, City Emergency Contact: 0300-XXXXXXX Username: login name

Password: password

Add User

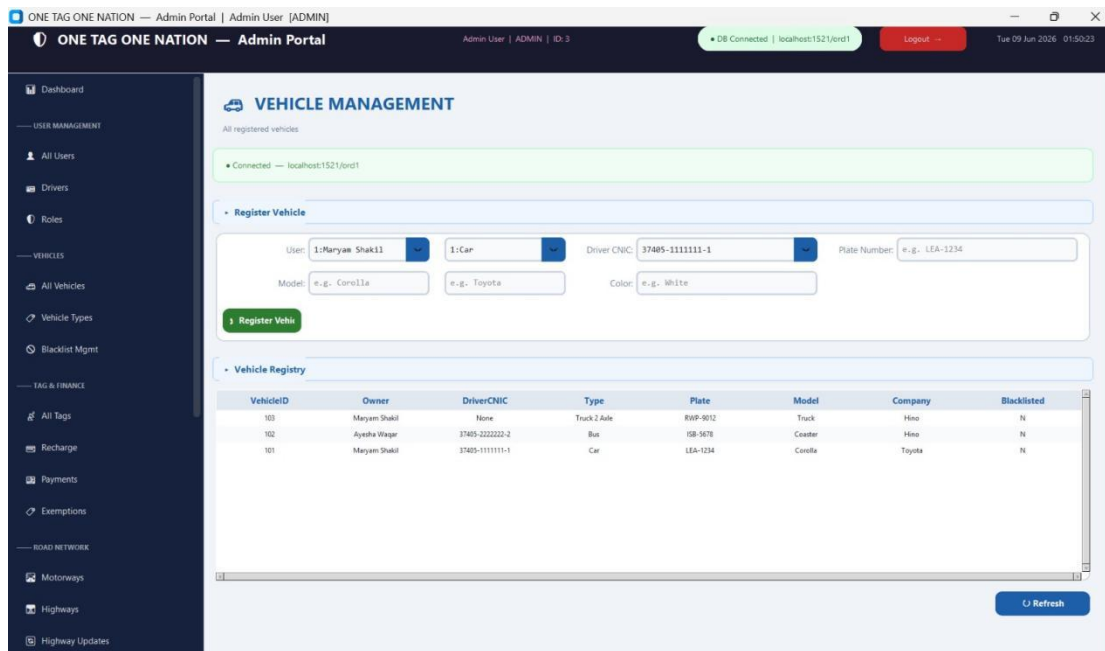
User List

UserID	Name	CNIC	Phone	Province	Email	Active	Username
3	Admin User	37405-9999999-9	03451234567	ICT	admin@etn.gov.pk	Y	admin
2	Ayesha Waqar	37405-7634021-2	01110234567	ICT	ayesha@gmail.com	Y	ayesha
1	Maryam Shakil	37405-1234567-1	03001234567	Punjab	maryam@gmail.com	Y	maryam

Refresh Toggle Active

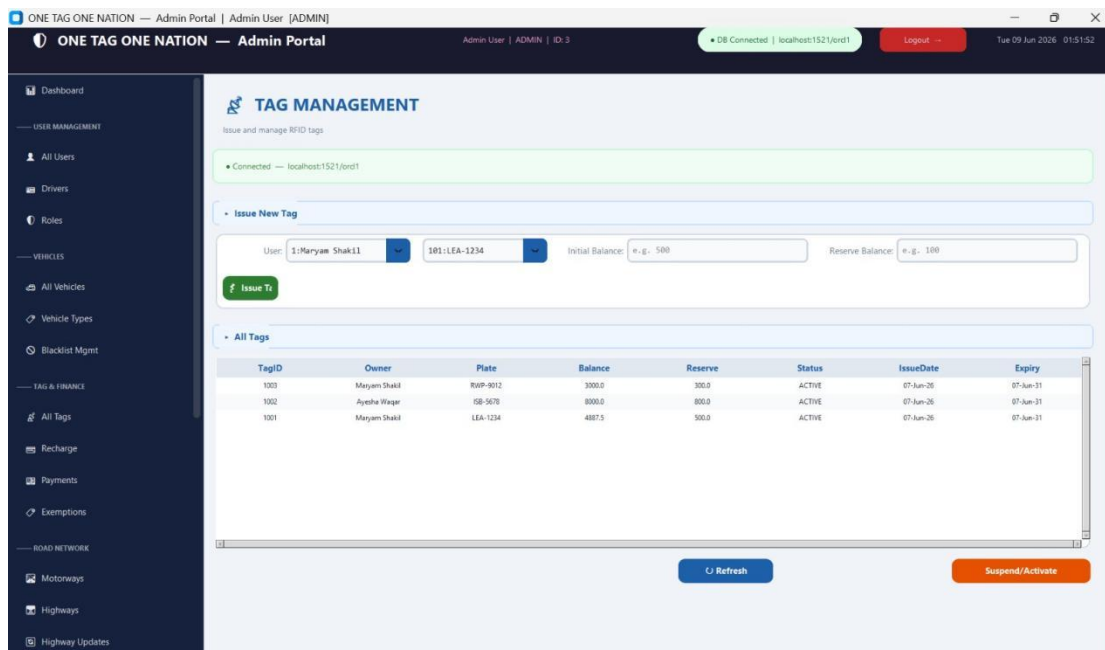
5.3 Vehicle Management Module

Admin can monitor all registered vehicles in system.



5.4 RFID Management Module

Admin can assign or remove RFID tags.



5.5 Toll Plaza & Motorway Management

Admin manages:

Toll plazas

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2026 01:53:08

All Users

Drivers

Roles

VEHICLES

All Vehicles

Vehicle Types

Blacklist Mgmt

TAG & FINANCE

All Tags

Recharge

Payments

Exemptions

ROAD NETWORK

Motorways

Highways

Highway Updates

Toll Plazas

Terminals

TOLL PLAZA MANAGEMENT

Add and manage toll plazas

Connected — localhost:1521/orcl1

Add Toll Plaza

Highway:1:MI North Section1:Normal TollPlaza Name:e.g., Kala Shah KakuLocation:e.g., KSK, Lahore

Province:Punjabe.g., 25.5Exit No:e.g., E-5

Add Plaza

Plaza List

PlazaID	Name	Highway	Province	Location	MileNo	ExitNo
1	Islamabad Toll Plaza	M1 North Section	ICT	Islamabad	0	E1
2	Burhan Toll Plaza	M1 North Section	Punjab	Burhan	45	E5
3	Bhera Toll Plaza	M2 Main Section	Punjab	Bhera	120	E12
5	sahawal	M1 North Section	—	sahawal	—	—

Refresh

Motorways

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2026 01:54:04

All Users

Drivers

Roles

VEHICLES

All Vehicles

Vehicle Types

Blacklist Mgmt

TAG & FINANCE

All Tags

Recharge

Payments

Exemptions

ROAD NETWORK

Motorways

Highways

Highway Updates

Toll Plazas

Terminals

MOTORWAY MANAGEMENT

National motorway network

Connected — localhost:1521/orcl1

Add Motorway

Motorway Name:e.g., M-2e.g., LahoreEnd City:e.g., IslamabadTotal Distance Km:e.g., 367

Add Motorway

Motorway List

MotorwayID	Name	Start	End	Distance(Km)
1	M1 Motorway	Islamabad	Peshawar	155.0
2	M2 Motorway	Lahore	Islamabad	367.0

Refresh

Highways

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2026 01:57:00

Payments

Exemptions

ROAD NETWORK

Motorways

Highways

Highway Updates

Toll Plazas

Terminals

Services

OPERATIONS

All Transactions

Employees

MONITORING

Cameras

Lanes

Alerts

All Feedback

Admin Log

HIGHWAY MANAGEMENT

Highway sections per motorway

Connected — localhost:1521/orcl1

Add Highway

Motorway: 1:M1 Motorway

e.g. W-2 Section A

Section Name: e.g. Lahore-Sheikhupura

Direction: BOTH

Start Location: e.g. Lahore

e.g. Pindi Bhattian

Distance (Km): e.g. 112

Add Highm

Highway List

HwyID	Motorway	Name	Section	Direction	Distance
1	M1 Motorway	M1 North Section	Islamabad Faisalpur	Northbound	135.0
2	M2 Motorway	M2 Main Section	Lahore-Islamabad	Southbound	367.0

Refresh

Terminals

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2026 01:57:26

Payments

Exemptions

ROAD NETWORK

Motorways

Highways

Highway Updates

Toll Plazas

Terminals

Services

OPERATIONS

All Transactions

Employees

MONITORING

Cameras

Lanes

Alerts

All Feedback

Admin Log

TERMINAL MANAGEMENT

IN/OUT terminals at toll plazas

Connected — localhost:1521/orcl1

Add Terminal

Plaza: 1:Islamabad Toll Plaza

e.g. KSK Entry 1

Type: IN

Mile No: e.g. 25.5

Add Termin

Terminal List

TerminalID	Plaza	Name	Type	MileNo	Status
1	Islamabad Toll Plaza	Islamabad Entry	IN	0	ACTIVE
2	Islamabad Toll Plaza	Islamabad Exit	OUT	25	ACTIVE
3	Burhan Toll Plaza	Burhan Entry	IN	45	ACTIVE
4	Burhan Toll Plaza	Burhan Exit	OUT	70	ACTIVE

Refresh

Other Services

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2026 02:00:04

Payments

Exemptions

ROAD NETWORK

Motorways

Highways

Highway Updates

Toll Plazas

Terminals

Services

OPERATIONS

All Transactions

Employees

MONITORING

Cameras

Lanes

Alerts

All Feedback

Admin Log

SERVICE MANAGEMENT

Toll service types and rates

Connected — localhost:1521/orcl1

Add Service

Service Name: Rate Per Km: Base Rate:

Surcharge Rate:

Add Servi

Service List

ServiceID	Name	Type	Rate/Km	BaseRate	Surcharge	EffectiveFrom
1	Normal Toll	STANDARD	5	5	0	07-Jun-26
2	Peak Hour Toll	PEAK	6	5	1	07-Jun-26
4	normal toll	—	10	—	0	07-Jun-26

Refresh

5.6 Transaction Monitoring Module

Admin can view all toll deductions in real time.

ONE TAG ONE NATION — Admin Portal | Admin User [ADMIN]

Admin User | ADMIN | ID: 3

DB Connected | localhost:1521/orcl1

Logout

Tue 09 Jun 2026 02:01:26

Payments

Exemptions

ROAD NETWORK

Motorways

Highways

Highway Updates

Toll Plazas

Terminals

Services

OPERATIONS

All Transactions

Employees

MONITORING

Cameras

Lanes

Alerts

All Feedback

Admin Log

ALL TRANSACTIONS

Full transaction ledger — admin view

Connected — localhost:1521/orcl1

Full Transaction Log

TransID	TagID	Plate	Plaza	EntryTerm	ExitTerm	Dist(Km)	Rate	Base	Surcharge	Discount	Total(Rs)	Date
21	1001	LEA-1234	Islamabad Toll Plaza	Islamabad Entry	Islamabad Exit	25.0	5.0	125.0	50.0	62.5	112.5	08-Jun-26 21:56
2	1002	ISB-5678	Burhan Toll Plaza	Burhan Entry	Burhan Exit	25.0	8.0	200.0	0.0	0.0	200.0	06-Jun-26 13:17
1	1001	LEA-1234	Islamabad Toll Plaza	Islamabad Entry	Islamabad Exit	25.0	5.0	125.0	0.0	0.0	125.0	07-Jun-26 13:17

Refresh

5.7 Reports Module

Admin can generate system reports:

Summary KPIs



Transaction Report

• Transaction Report — Distance × Rate Formula

TxnID	TagID	Plate	Plaza	EntryTerm	ExitTerm	Dist(Km)	Rate/Km	BaseToll	Surcharge	Discount	Total(Rs)	Date
21	1001	LEA-1234	Islamabad Toll Plaza	Islamabad Entry	Islamabad Exit	25.0	5.0	125.0	50.0	62.5	112.5	08-Jun-26 21:56
2	1002	ISB-5678	Burhan Toll Plaza	Burhan Entry	Burhan Exit	25.0	8.0	200.0	0.0	0.0	200.0	06-Jun-26 13:17
1	1001	LEA-1234	Islamabad Toll Plaza	Islamabad Entry	Islamabad Exit	25.0	5.0	125.0	0.0	0.0	125.0	07-Jun-26 13:17

Revenue reports

• Revenue by Plaza — GROUP BY

PlazaName	Transactions	Total Revenue (Rs)	Avg Toll (Rs)
Islamabad Toll Plaza	2	237.5	118.75
Burhan Toll Plaza	1	200	200

Tag Balance Report

• Tag Balance Report — LOW BALANCE WARNING

TagID	Owner	Plate	VehicleType	Balance	ReserveBalance	Status
1003	Maryam Shakil	RWP-9012	Truck 2 Axle	3000.0	300.0	ACTIVE
1001	Maryam Shakil	LEA-1234	Car	4887.5	500.0	ACTIVE
1002	Ayesha Waqar	ISB-5678	Bus	8000.0	800.0	ACTIVE

User Role reports

• User Roles — JOIN Example

Name	Username	Role	AssignedDate
Maryam Shakil	maryam	CUSTOMER	2026-06-07
Ayesha Waqar	ayesha	CUSTOMER	2026-06-07
Admin User	admin	ADMIN	2026-06-07

5.8 Feedback Management Module

Admin can view user feedback.


USER FEEDBACK

All user feedback and ratings

Connected — localhost:1521/ord1

Feedback Log

FeedbackID	UserID	Name	Rating	Comments
3	1	Maryam Shakil	5	smooth working
2	2	Ayesha Waqar	4	Fast and smooth passage
1	1	Maryam Shakil	5	Excellent Toll Service

Refresh

5.9 Admin Audit Log Module

Admin can view admin audit log.


ADMIN AUDIT LOG

All admin actions — tamper-evident trail

Connected — localhost:1521/ord1

Admin Action Log

LogID	AdminID	AdminName	ActionType	Detail	Timestamp
2	3	Admin User	RATE_UPDATE	Updated toll rates for M1	07-Jun-26 13:17:35
1	3	Admin User	LOGIN	Admin logged into system	07-Jun-26 13:17:35

Refresh

5.10 Camera System Module

Admins can monitor cameras, add cameras and see cameras list.

CAMERA SYSTEM

Surveillance cameras at plazas

• Connected — localhost:1521/ord1

• Add Camera

Plaza: 1: Islamabad Toll Plaza

Location: e.g. Entry Gate Left

Add Camera

• Camera List

CameraID	Plaza	Location	Status
3	Burhan Toll Plaza	Entry Gate	ACTIVE
2	Islamabad Toll Plaza	Exit Gate	ACTIVE
1	Islamabad Toll Plaza	Entry Gate	ACTIVE

Refresh

Toggle Status

6. SYSTEM WORKFLOW

User/Admin opens application

Login or Signup

Role-based dashboard opens

User performs actions (vehicles, recharge, etc.)

Admin monitors and controls system

All data stored in Oracle database

Transactions updated in real-time

9. CONCLUSION

The Toll Management System successfully provides a dual-role platform for users and administrators. The system improves efficiency, reduces manual workload, and ensures accurate toll tracking with secure database integration.

The separation of user and admin modules ensures clarity, scalability, and better system management.